Research on Courses Relationship Model Based on Bayesian Networks

Jianming Huang
Computer Center
Kunming University of Science and Technology
Kunming 650093, China
hjmkm@163.com

Jiaoli Fang
Computer Center
Kunming University of Science and Technology
Kunming 650093, China
fangjiaoli@163.com

***Abstract*—University courses are not isolated settings. There is a certain link among them. This paper presents a construction method for Bayesian networks of university courses relationship, which use the examination results of students' courses as data sample. The undirected graph was constructed with structure learning algorithm based on information theory, and its edges were oriented according to the time order of courses opening. Such the Bayesian network of courses dependence relationship was obtained, and its condition probability table was learned by mathematical statistics method. This model had represented the dependence relationship of courses intuitively, and the condition probability table quantified tightness of relationship. It plays guidance role to the setting and arrangement of university courses, and it has the prediction ability to the students' achievement.**

Keywords-Bayesian networks; information theory; undirected graph; structure learning; condition probability table

I. INTRODUCTION

Bayesian network [1, 2] is a directed acyclic graph (DAG) where each node represents a random variable, it is a model based on theory of probability and graph used to express and infer uncertain knowledge. It has a solid theoretical foundation, the natural express to knowledge structure. In recent years, Bayesian networks have been widely used in various fields of medical diagnosis, data mining, pattern recognition, decision support etc [3, 4].

Bayesian networks can be expressed as a triple (*N, E, P)*. The *N* is a group of nodes set, and $N = \{x_1, x_2, ..., x_n\}$, each node represents a variable. The *E* is a group directed edges set, and $E = \{\langle x_i, x_j \rangle \mid x_i \neq x_j \ and \ x_i, x_j \in N\}$. Each edge $\langle x_i, x_j \rangle$ represents that between variables x_i, x_j have a direct causal link. The *P* is a group condition probability set, and $P = \{P(x_i \mid parents(x_i))\}$ is conditional probability function set of each node. The full joint distribution of Bayesian networks is given by the product:

$$P(x_1, x_2, ..., x_n) = \prod_{i=1}^{n} P\left(x_i \mid parents(x_i)\right) \qquad (1)$$

Each major has the corresponding course system in university. The courses are not isolated settings, and there is a certain relationship among them. A Bayesian network can be obtained from experts, or by learning from data samples [5, 6]. This paper presents a construction method for Bayesian network of university courses relationship, which uses the examination results of students' courses as data sample. first undirected graph was constructed by structure learning algorithm based on information theory, and its edges were oriented according to the time order of courses opening, then its network parameters are learned, finally the Bayesian network of courses dependence relationship was obtained. This model has represented the dependence relationship of courses intuitively, and the condition probability table has quantified tightness of the relationship. It plays a guidance role to the setting and arrangement of university courses. It can predict the students' achievement easily, and has the good application value.

II. AN UNDIRECTED GRAPH CONSTRUCTION ALGORITHM BASED ON INFORMATION THEORY

An undirected graph is a graph in which the nodes are connected by undirected arcs. An undirected graph is an ordered pair *G*=<*E*, *V*>, with the following properties:

1. The first component, *E*, is a finite, non-empty set. The elements of *E* are called the vertices of *G*.

2. The second component, *V*, is a finite set of sets. Each element of *V* is a set that is comprised of exactly two (distinct) vertices. The elements of *V* are called the edges of *G*.

Construction of undirected graph does not need to determine the direction of edges, and it is easier than learning Bayesian network directly.

DEFINITION 1: Let *U* be a finite variables set, *U* has any three disjoint subsets *X*, *Y*, and *Z*. *I*（*X*, *Z*, *Y*） represents *X* and *Y* be conditional independence given *Z***,** dependent model is defined as a set of condition independence.

DEFINITION 2: Under known variable *Z* condition, definition conditional mutual information *I*（*X*, *Y* | *Z*）of variables *X* and *Y* :

$$I(X,Y \mid Z) = \sum_{I=1}^{I}\sum_{J=1}^{J}\sum_{K=1}^{K} p(x_i, y_j, z_k) \log \frac{p(x_i, y_j \mid z_k)}{p(x_i \mid z_k) p(y_j \mid z_k)} \quad (2)$$

To set a threshold value (one small positive real number), if $I(X,Y \mid Z) \leq \varepsilon$, while given *Z*, *X* and *Y* condition independence, that is $I(X,Z,Y)$ tenable. Otherwise, while given *Z, X* and *Y* is not condition independence.

978-0-7695-3645-3/09 $25.00 © 2009 IEEE
DOI 10.1109/CINC.2009.222

In definition conditional mutual information, probability $P(x_i, y_j, z_k)$, $P(x_i, y_j | z_k)$, $P(x_i | z_k)$ and $P(y_j | z_k)$ can be obtained from samples by statistical calculation.

We regard data item of samples that combination of two as （α，β）, test $I(\alpha, U-\alpha-\beta, \beta)$, if $I(\alpha, U-\alpha-\beta, \beta)$ not tenable, then connect the two nodes in the undirected graph, as an edge of undirected graph.

Let U is a set of all data items in sample.

The algorithm described as follows:

1) Input the sample of contains N data items, and a threshold value ε ;
2) In the N data items, test conditional mutual information I（X, Y | Z） in any two data items X and Y successively, among which $Z=U-X-Y$;
3) If $I(X, Y | Z) > \varepsilon$, then has an edge between X and Y; otherwise, they have not directly connected edge;
4) Output the undirected graph of contains all the edges.

Undirected graph can not reflect relationship of causal and dependent between nodes, as soon as edges of undirected graph were oriented, then can obtains the corresponding Bayesian network. Usually directional arc methods of Bayesian networks mainly include collision identify and scoring & search two methods [7]. Collision identify is a method that arc is oriented by means of V structure. The so-called V structure is a network local structure that shape like $A \rightarrow C \leftarrow B$. The basic idea of collision identify is: if two not adjacency nodes A and B, there is undirected edge to connection between C node and them, and C is not cut set of A and B, then C is a collision nodes, Then determine the direction of edge for $A \rightarrow C \leftarrow B$, the structure is called V structure of Bayesian networks.

In this paper, directional arc method of Bayesian networks is that according to sequencing of node event occurred.

III. CONSTRUCTION OF UNIVERSITY COURSES MODEL BASED ON BAYESIAN NETWORKS

This paper use the examination results of students' courses as data sample, after cleaning, discrete processing and statistical analysis, undirected graph was constructed with structure learning algorithm based on information theory, and its edges were oriented according to the time order of courses opening. Finally, the Bayesian network of courses dependence relationship was obtained.

In this paper, the data sample collected from a university, 2000-2004 grade students of "mechanical design manufacturing and automation" major, a total of more than 1600 students. We selected seven main courses in the major that include: “higher mathematics”, “college physics”, “theoretical mechanics”, “mechanical principle”, “material mechanics”, “mechanical design”, “machinery manufacturing”, and use the examination results of students' courses as data sample. In order to conveniently express, this paper used C1-C7 to represent above seven courses successively.

We collected more than 1600 students’ examination results of seven courses as table Ⅰ.

TABLE I. STUDENTS’ EXAMINATION RESULTS OF COURSES

Student ID	Courses Name						
	C1	C2	C3	C4	C5	C6	C7
0401001	65	60	51	83	63	87	76
0401002	70	64	73	81	58	64	75
0401003	80	78	81	68	69	72	79
0401004	69	65	73	81	74	69	71
0401005	56	64	68	72	69	73	62
…	…	…	…	…	…	…	…

Then these data are processed by means of discrete method. Specific approach is: each students’ examination results of courses is divided into two paragraphs, if greater than or equal to 70, it is denoted by “high”; if less than 70, it is denoted by “low”. The results are such as table Ⅱ.

TABLE II. STUDENTS’ SCORE OF COURSES EVALUATION

Student ID	Courses Name						
	C1	C2	C3	C4	C5	C6	C7
0401001	low	low	low	high	low	high	high
0401002	high	low	high	high	low	low	high
0401003	high	high	high	low	low	high	high
0401004	low	low	high	high	high	low	high
0401005	low	low	low	high	low	high	low
…	…	…	…	…	…	…	…

Pay attention to after data discretization, some courses of different students are the same score of evaluation in table Ⅱ. Next step, we deal these data by projected method in table Ⅱ, and obtained the repeat number of tuple. The results are such as table Ⅲ.

TABLE III. STUDENTS’ SCORE SAMPLE STATISTICS

Courses Name							Number
C1	C2	C3	C4	C5	C6	C7	
low	low	low	low	low	low	low	213
low	low	low	low	low	low	high	30
low	low	low	low	low	high	low	85
low	low	low	low	low	high	high	59
…	…	…	…	…	…	…	…
high	high	high	high	high	high	high	229

As can be seen from table Ⅲ, in a total of 1604 students, there are 213 people who have got “low” in 7 courses, and there are 229 people who have got “high”. (The length limit, do not listed all data in table Ⅲ).

Subsequently, we constructed undirected graph by means of structure learning algorithm based on information theory, and with table Ⅲ as data sample. The algorithm used threshold value ε =0.02, obtained undirected graph model which includes 9 edges, that is Fig. 1.

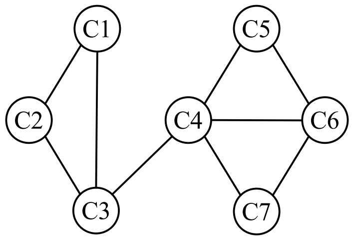

Figure 1. Undirected graph model of courses relationship

Next step, its edges were oriented according to the time order of courses opening, such the Bayesian network of courses dependence relationship was obtained. The method of oriented edges: if any two nodes (courses) have no undirected edge to connect in network, then first occurrence event point to later occurrence event (namely first establishment courses point to later establishment courses). Table Ⅳ is the time order of 7 courses opening.

TABLE IV. COURSES OPENING TIME

Courses name	Opening time
higher mathematics	The first semester
college physics	The second semester
theoretical mechanics	The third semester
material mechanics	The fourth semester
mechanical principle	The fourth semester
mechanical design	The fifth semester
machinery manufacturing	The sixth semester

According to the time order of courses opening in table Ⅳ, we give direction of edges for Fig. 1, such the Bayesian network of courses dependence relationship was obtained, and that is Fig. 2.

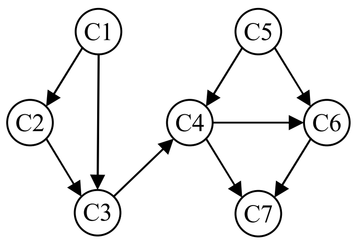

Figure 2. Bayesian network of courses dependence relationship

From the Fig. 2 we can see that the seven courses are divided into two modules. The right part is module of basic courses, including the "higher mathematics", "college physics", "theoretical mechanics" and other courses; and the left part is module of professional courses, including the "mechanical principle", "material mechanics", "mechanical design", "machinery manufacturing" and other courses. The entire network model includes 9 edges. There is no edges connected among some courses, this is because the "information" is weak among them, it is not closely linked than foregoing 9 edges. Certainly, want to change number of edges in the network, as long as the threshold value ε can make appropriate adjustments in algorithm.

IV. LEARNING CONDITION PROBABILITY TABLE

A complete Bayesian network has not only the network structure, but also should include its condition probability table. We have obtained its condition probability table by means of mathematical statistics method from table Ⅲ.

Such as in Fig. 2, "higher mathematics" and "material mechanics" nodes have no parent nodes, then the marginal probability of "higher mathematics" node as follows:

$$P(C1\text{=high})=\frac{\text{number of "}C1\text{=high"}}{\text{total number}}=\frac{612}{1604}\approx 0.38$$

Correspondingly,

$$P(C1\text{=low})=1\text{-}P(C1\text{=high})=1\text{-}0.38=0.62$$

For "college physics" node, its parent node is "higher mathematics", so its condition probability is:

$$P(C2\text{=high}\mid C1\text{=high})=\frac{\text{number of "}C1\text{ and }C2\text{=high"}}{\text{number of "}C1\text{=high"}}\approx 0.75$$

$$P(C2\text{=low}\mid C1\text{=high})=1\text{-}P(C2\text{=high}\mid C1\text{=high})=0.25$$

By the same way,

$$P(C2\text{=high}\mid C1\text{=low})=\frac{\text{number of "}C1\text{ and }C2\text{=low"}}{\text{number of "}C1\text{=low"}}\approx 0.22$$

By the same method, we can obtain condition probability of other nodes. Fig. 3 is that we have obtained condition probability table by means of statistical learning method.

P (C1=high) =0.38
P (C2=high| C1=high) =0.75
P (C2=high| C1=low) =0.22
P (C3=high | C1=high, C2=high) =0.71
P (C3=high | C1=high, C2=low) =0.23
P (C3=high | C1=low, C2=high) =0.35
P (C3=high | C1=low, C2=low) =0.11
P (C4=high | C3=high, C5=high) =0.88
P (C4=high | C3=high, C5=low) =0.56
P (C4=high | C3=low, C5=high) =0.59
P (C4=high | C3=low, C5=low) =0.27
P (C5=high) =0.5
P (C6=high | C4=high, C5=high) =0.94
P (C6=high | C4=high, C5=low) =0.75
P (C6=high | C4=low, C5=high) =0.64
P (C6=high | C4=low, C5=low) =0.41
P (C7=high | C4=high, C6=high) =0.75
P (C7=high | C4=high, C6=low) =0.34
P (C7=high | C4=low, C6=high) =0.45
P (C7=high | C4=low, C6=low) =0.14

Figure 3. Condition probability table

Condition probability table is obtained, as long as by means of Bayesian networks inference, that can predict students' achievement.

V. CONCLUSIONS

This paper presents a construction method for courses relationship Bayesian networks, which use the examination results of students' courses as data sample, to mining dependence relationship among courses, construct Bayesian network of university courses relationship. This model had represented the dependence relationship of courses intuitively. It plays guidance and auxiliary role to the setting and arrangement of university courses. It has good application value. The result showed that construction method of model is feasible and effective. Of course, because the data sample is not large enough in this paper, therefore, the model did not include more courses. The network model has constructed, next step students' achievement can predicted by means of this model.

REFERENCES

[1] J. Pear, "Probalilistic reasoning in intelligent systems : Networks of plausible inference," Morgan Kaufmann Publishers Inc., San Mateo, CA, 1988.

[2] D. Heckerman, "A Tutorial on Learning With Bayesian Networks," Microsoft Research, Technical Report: MSR-TR-95-06, 1995.

[3] D. Heckerman, "Bayesian networks for data mining," Data Mining and Knowledge Discovery, vol. 1, March. 1997, pp. 79-119.

[4] D. Heckerman, A. Mamdani and M. P. Wellman, "Real-world applications of Bayesian networks," Communications of the ACM, vol. 38, March. 1995, pp. 24-26.

[5] G. F. Cooper and E. Herskovits, "A Bayesian Method for the Induction of Probabilistic Networks from Data," Machine Learning, vol. 9, September. 1992， pp. 309-347.

[6] J. Cheng, D. Bell and W. Liu, "Learning Bayesian networks from data: An efficient approach based on information theory," Artificial Intelligence, vol. 137, January. 2002, pp. 43-90.

[7] Wang Shuang-Cheng and Yuan Sen-Miao, "Research on Learning Bayesian Networks Structure with Missing Data," Journal of Software, vol. 15, July. 2004, pp. 1042-1048.